

When AI Memory Changes What You Told It

Six controlled experiments on memory integrity, interpretation, and correctability in adaptive AI

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Correct memory does not guarantee correct understanding. An AI can preserve what a person said while its interpretation of that person becomes unsupported, no longer valid, or too broad.

Abstract. Persistent AI systems increasingly build representations of users across interactions. This creates a problem that is easy to miss: preserving the underlying memory does not guarantee that the system’s interpretation of the person remains faithful. Across six controlled synthetic-user experiments, I tested successive approaches to memory updating, atomic revision, immutable evidence, evidence-grounded interpretation, verification, and decomposed verification. Each approach addressed part of the problem, but none solved the full chain. The final experiment exposed a particularly sharp tradeoff: a decomposed verifier detected all eight deliberately corrupted interpretations, yet rejected all eight valid interpretations. The sequence motivates a narrower alignment problem I call Interpretation Correctability: keeping personalized AI beliefs useful while ensuring they remain grounded, current, bounded, calibrated, and revisable.

1. The problem is not only memory

Long-term AI memory is usually discussed as a storage and retrieval problem: can the system remember relevant information and use it later? The experiments reported here began with that question but repeatedly encountered a different failure. The system could preserve information and still become wrong about the person represented by that information.

Consider a user who says she is waiting to join a research collaboration because she needs more evidence. A useful system may remember that statement. But it can still transform a local reason into a trait such as “she tends to hesitate before committing to research collaborations.” Later, if she says the evidence is sufficient and she is now waiting for budget approval, the historical statement remains true as history while no longer explaining the current delay. A memory system therefore needs more than recall. It needs to distinguish evidence from interpretation, historical truth from current validity, and local context from general preference.

Stage	Question	What became visible
Memory	Can structured semantic memory improve adaptation?	Memory maintenance itself can distort meaning.
Revision	Can atomic local updates prevent degradation?	Atomicity protects unrelated records, not necessarily the

		record being revised.
Evidence	Can immutable evidence preserve a correction path?	Yes, but evidence does not guarantee faithful interpretation.
Interpretation	Can grounding prevent unsupported beliefs?	It can help while interpretation and application still fail.
Verification	Can evidence verify and repair a corrupted interpretation?	A verifier can mistake plausibility, breadth, or stale evidence for support.
Calibration	Can explicit checks improve discrimination?	They can catch more corruptions while rejecting valid personalization.

2. Six experiments, one progressively narrower failure

Experiment 1: repeated correction

Four memory conditions were compared: no persistent memory, raw conversation history, repeatedly rewritten structured semantic memory, and a rolling summary. No architecture was universally superior. Structured memory sometimes personalized correctly, but repeated updating could alter, generalize, or remove supplied information. The result was not evidence that structured memory is inherently harmful; output limits and the specific rewriting implementation were important confounds. It did establish memory integrity during updating as a problem worth isolating.

Experiment 2: atomic revision

The next test separated holistic rewriting from atomic local revision, with a third condition adding provenance. Atomicity protected unrelated records from repeated rewriting, but targeted revisions still distorted meaning. Provenance preserved a more auditable historical record without preventing active semantic distortion. This suggested two distinct mechanisms: exposure degradation, caused by rewriting information that did not need to change, and revision degradation, caused while changing the record that actually did need an update.

Experiment 3: immutable evidence

Human statements were then preserved as immutable evidence while the AI maintained a separate mutable interpretation. The evidence remained intact and enabled partial reconstruction after deliberate corruption, but active interpretation fidelity did not improve overall: the mutable atomic condition scored 14/20 across checkpoints and the evidence-plus-interpretation condition 13/20. Recovery from evidence scored 15/20. The important result was conceptual: preserving what the human said can preserve a correction path without guaranteeing that the AI will reconstruct or apply it correctly.

Experiment 4: evidence-grounded interpretation

Explicit grounding improved rejection of several prespecified unsupported trait-level claims. At the final audit, the grounded condition rejected all three unsupported claims while retaining one supported personalization. Yet the same condition generated unsupported abstractions in its active interpretation

and sometimes failed to use valid personalization. Evidence integrity, interpretation integrity, contextual application, and behavior therefore had to be treated as separate layers.

Experiment 5: verification and repair

The next system was given immutable, versioned evidence and asked to verify deliberately corrupted interpretations before repairing them. It classified every case as SUPPORTED, including all four corruptions. No repair occurred. The failure happened before the recovery mechanism: the verifier treated plausible, overgeneralized, unsupported, or temporally obsolete interpretations as justified by the evidence.

Experiment 6: decomposed verification

Verification was decomposed into entailment, temporal validity, and contextual scope. This fixed one problem and created another. The holistic verifier achieved 13/16 accuracy, caught 5/8 corruptions, and preserved 8/8 valid interpretations. The decomposed verifier caught 8/8 corruptions but preserved 0/8 valid interpretations, for 8/16 accuracy. Mechanical composition produced the same outcome. More explicit skepticism increased sensitivity while destroying calibration.

3. What the sequence actually supports

The strongest result is not that one memory architecture won. It did not. The sequence instead separates failure modes that are often collapsed into “memory quality.” A system can store the right evidence and derive the wrong interpretation. It can hold a reasonable interpretation that is no longer current. It can take a preference that is valid in one context and generalize it into a trait. It can retrieve the right information and still apply it incorrectly. Finally, a verifier can become so cautious that it refuses to use information that is directly supported.

This creates a calibration problem: when does an adaptive AI have enough evidence to believe something about a person, with what confidence, in which context, and until when?

The experiments also show why behavioral evaluation alone is insufficient. Generic reasoning can sometimes produce a good answer despite corrupted memory. Conversely, a correct stored representation can be misapplied downstream. Memory, interpretation, verification, application, and behavior need separate measurements if we want to know where an adaptive system failed.

4. Interpretation Correctability

I use Interpretation Correctability as a working name for the capability exposed by this sequence: the ability of an adaptive AI to maintain useful inferences about a person while preserving the distinction between human-supplied evidence and machine-generated interpretation, and to detect, revise, or withdraw interpretations when they become unsupported, no longer valid, contradicted, or contextually inappropriate.

The term is intentionally narrower than a claim that a new memory architecture has been solved. Existing research already addresses evolving preferences, append-only temporal memory, memory update gaps, representational accuracy, and when personalization should be withheld. The candidate contribution here is the justification relationship itself: not merely what the system stores, but why a personalized interpretation is warranted by the human evidence available to it.

Human evidence -> mutable interpretation -> justification state -> temporal status -> contextual scope -> application decision -> behavior

This architecture is motivated by the failures, not validated by them. The experiments do not yet show that explicitly representing all of these layers will solve the problem. In particular, Experiment 6 warns that simply adding more checks can turn appropriate caution into systematic over-rejection.

5. Why this matters beyond personal assistants

The problem becomes more consequential as AI systems persist across time and adapt to individuals. In enterprise agents, a stale interpretation of a user's priorities can affect recommendations and workflows. In human-AI collaboration, a bounded preference can be transformed into an unjustified general rule. In Physical AI and robotics, systems may eventually combine interaction history with contextual, behavioral, or physiological signals. Better memory retrieval would help, but retrieval alone cannot determine whether a machine-generated belief is justified or safe to apply.

This also suggests a useful distinction for semantic memory systems. Similarity retrieval asks which memories resemble the current query. Correctability-oriented retrieval would additionally seek evidence that contradicts, supersedes, qualifies, or bounds the current interpretation. That is a testable direction, but it remains an open hypothesis rather than a result of the six experiments.

6. What remains unresolved

These are small controlled experiments, not a benchmark of real-world prevalence. They use one synthetic user, a small set of hand-designed cases, predominantly one open model family, deterministic single runs, and manual scoring in several experiments. Early runs include implementation confounds. Experiment 5 contains one metadata deviation that made a temporal case easier than intended. No real-user longitudinal deployment or independent replication has yet been performed.

Those limitations are also why the next step is not another internally designed experiment. The more valuable step is external replication. The notebooks, frozen later protocols, negative results, scoring rules, and implementation failures are being packaged so that another researcher or builder can rerun the tests on a different model or memory system without first accepting the framing.

Conclusion

The research began by asking which memory architecture adapts best to human correction. Six experiments later, the harder question appears to be epistemic rather than purely architectural: when is an AI justified in believing something about a person? A useful adaptive system should be able to learn from us without silently turning local evidence into enduring traits, historical reasons into current explanations, or caution into refusal to personalize at all.

The alignment target is simple to state and difficult to implement: use what you have legitimately learned about me, know why you believe it, know where it applies, and know when new evidence should change it.

Selected related work

- Banerjee, P., Moshtaghi, M., Subramanian, S., Misra, A., & Chadha, A. (2026). APEX-MEM: Agentic Semi-Structured Memory with Temporal Reasoning for Long-Term Conversational AI. ACL 2026.
- Gulaya, A. (2026). Beyond Recall: Behavioral Specification as an Interpretive Layer for AI Personalization. arXiv:2605.28969.
- Jiang, B., et al. (2026). PersonaMem-v3: Toward Omni-Platform Personal Intelligence for Holistic User Understanding, Recommendation, and Agentic Tasks. arXiv:2608.21381.
- Patel, V. (2026). Supersede: Diagnosing and Training the Memory-Update Gap in LLM Agents. arXiv:2606.27472.
- Sun, H., Zhang, Z., & Zeng, S. (2026). Preference-Aware Memory Update for Long-Term LLM Agents. Findings of ACL 2026.